



Generative Model of Suitable Meme Sentences for Images using AutoEncoder

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Outline

- Introduction
- What is meme?
- Conventional research
- Proposed method
- Experiments
- Conclusion

Introduction

Neural Networks have been used in various fields.

However, there are few researches using neural networks in joking.

- Laughing is an important factor for health.
- Covid-19 has created a demand for AI to provide laughter without fear of infection.

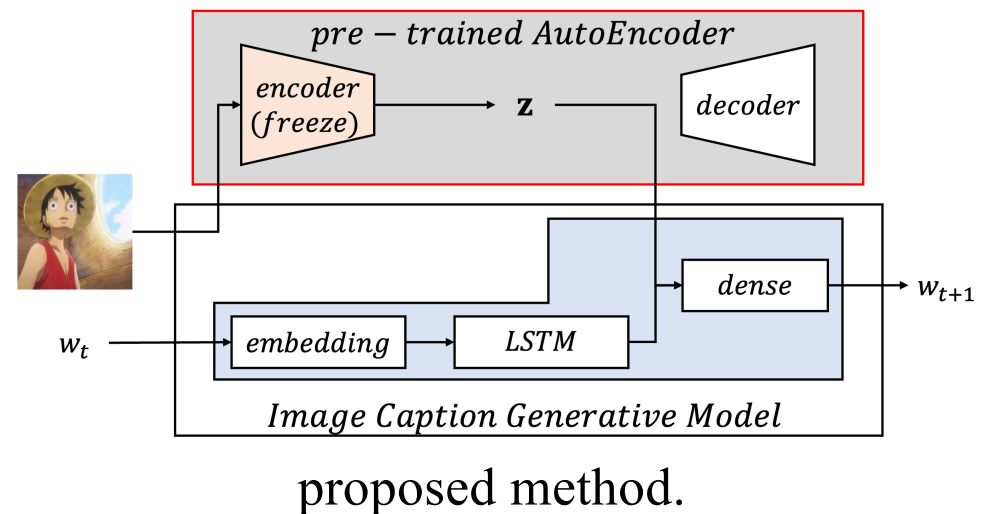
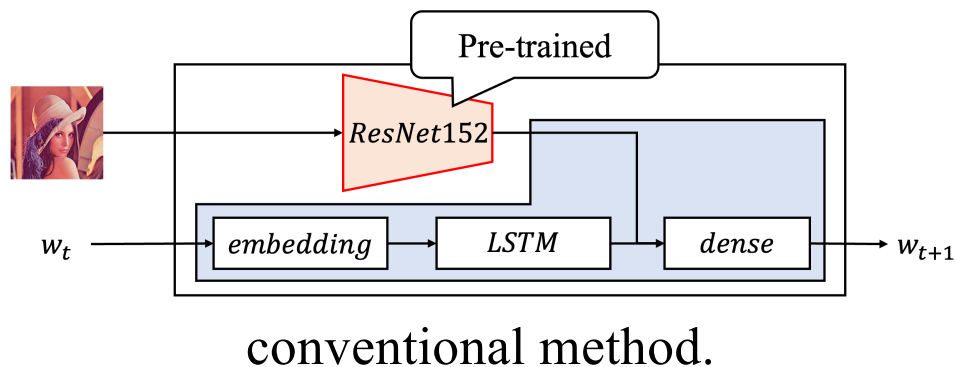


This research implements an AI that generates funny sentences for images.

Meme

Introduction

- Conventional method sometime generate unsuitable memes for unnatural images, such as animation or comics.
- Using AutoEncoder to deal with **domain divergence**.
- Proposed method trained better than conventional method.
- The proposed method has flexibility for model structure.



Meme[1]

- A meme is one of the cybercultures spreading and attracting on SNS.
- A meme is the pair of an image and a funny sentence suitable for the image.



Fig. 1. Memes generated by proposed method.

[1] Akhther, N.: Internet Memes as Form of Cultural Discourse: A Rhetorical Analysis on Facebook. PsyArXiv (2021).

Japanese meme : Bokete[2]

- A website that allows users to post Ogiri for images.
- Ogiri is a Japanese meme.
- 100 million Japanese memes and 7.3 million images have been posted.
- All images have **no label**.



bokete photo by ayumi3218 odai by ayumi3218

猫が運転してた
Cat was Driving.



bokete photo by わん odai by わん

AEDを持って来たわ!
I've got AED!



bokete photo by 頭頂砂漠 odai by 頭頂砂漠

車の鍵あったよー!!
I found the key for the car.

Fig. 2. Memes posted in Bokete.

[2] Omoroki Inc., Bokete, <https://bokete.jp>, Bokete, Last accessed 2023/10/5

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Conventional research : Neural Joking Machine[3]

- Scraping of **Bokete** website and collection of image and meme pairs.
- Image and meme pairs were trained on a image caption generative model.
- ResNet152, pre-trained on ImageNet, was used as an image encoder.
- The fun of the Memes generated by the model was evaluated in a questionnaire.



Fig. 3. Memes generated by Neural Joking Machine(NJM).

[3] Yoshida, K., et al.: Neural joking machine: Humorous image captioning. Proc. IEEE Computer Vision and Pattern Recognition Conference (CVPR) Language and Vision workshop (2018).

Conventional research : Neural Joking Machine

- Scraping of Bokete website and collection of image and meme pairs.
- Image and meme pairs were trained on an **image caption generative model**.
- **ResNet152, pre-trained on ImageNet**, was used as an image encoder.
- Transfer learning (freeze the parameters of ResNet152 while training).
- The fun of the memes generated by NJM was evaluated in a questionnaire.

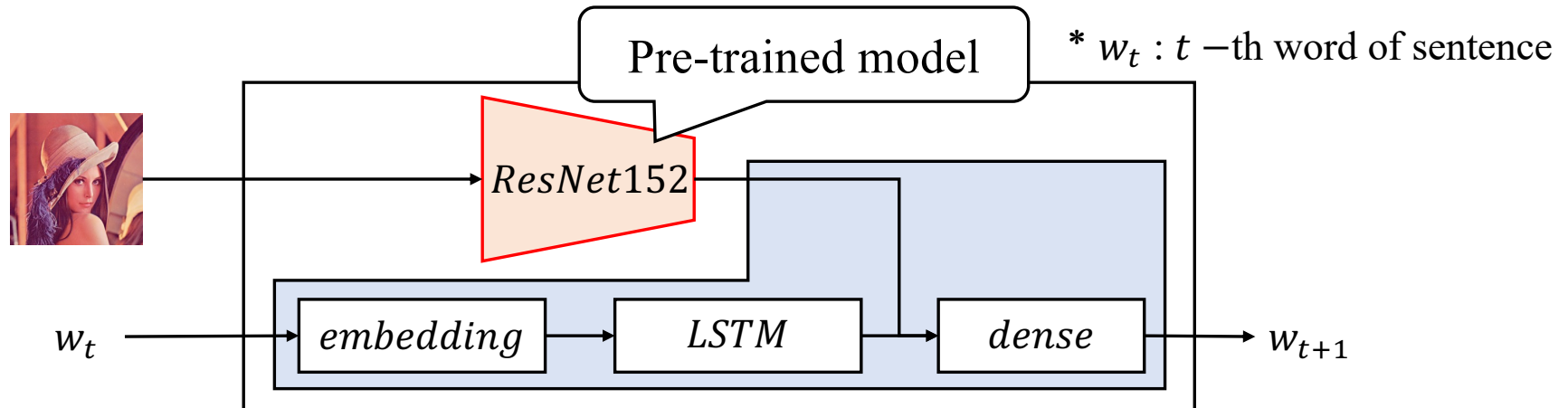


Fig. 4. The structure of NJM.

Problem

- NJM sometime generate memes unsuitable for images, because of **domain divergence**.



Fig. 5. Memes generated by NJM.

Domain divergence

- Domain divergence is a serious problem that causes a decrease in learning accuracy.

* only natural images

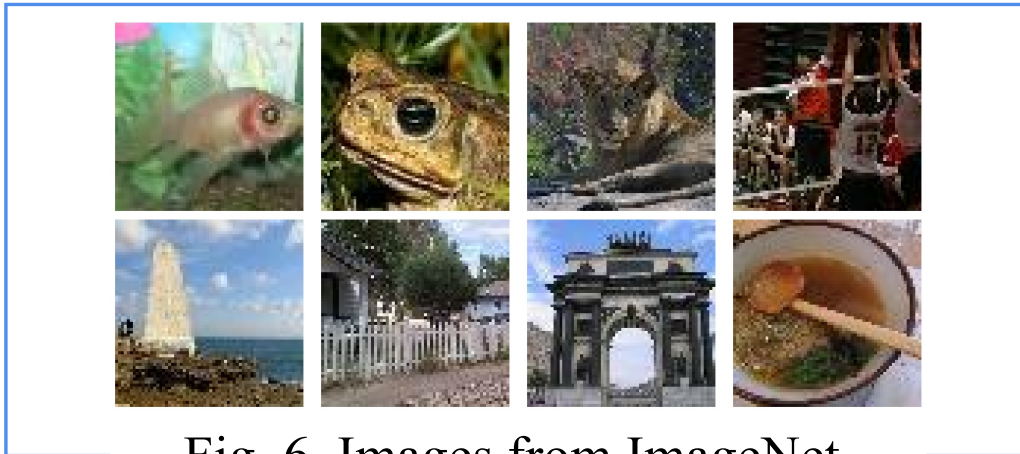


Fig. 6. Images from ImageNet.

* natural images, animations, comics, etc

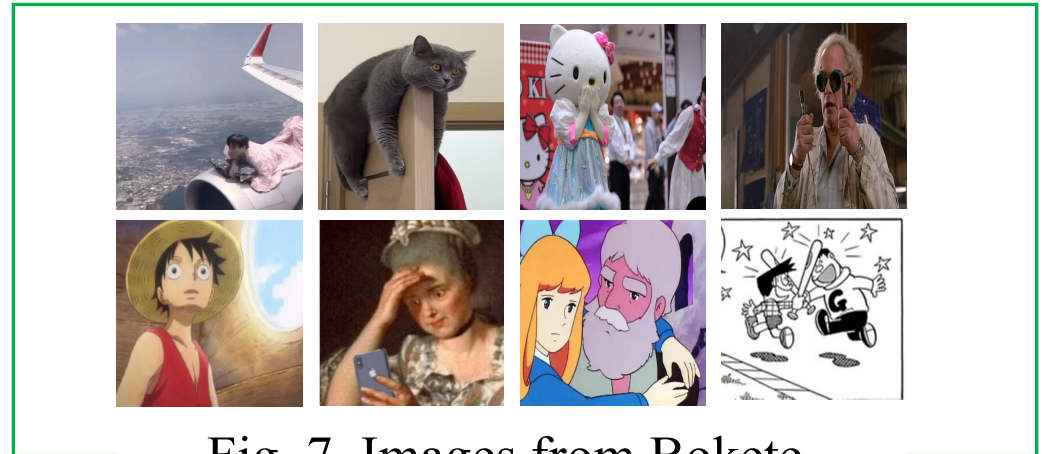


Fig. 7. Images from Bokete.

- Bokete images have **domain divergence** from ImageNet images.
- The purpose of this research is to deal with domain divergence and to generate memes with high accuracy.

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Proposed method

- Using the encoder of the **AutoEncoder**, pre-trained on Bokete images, as an image encoder of an image caption generative model.

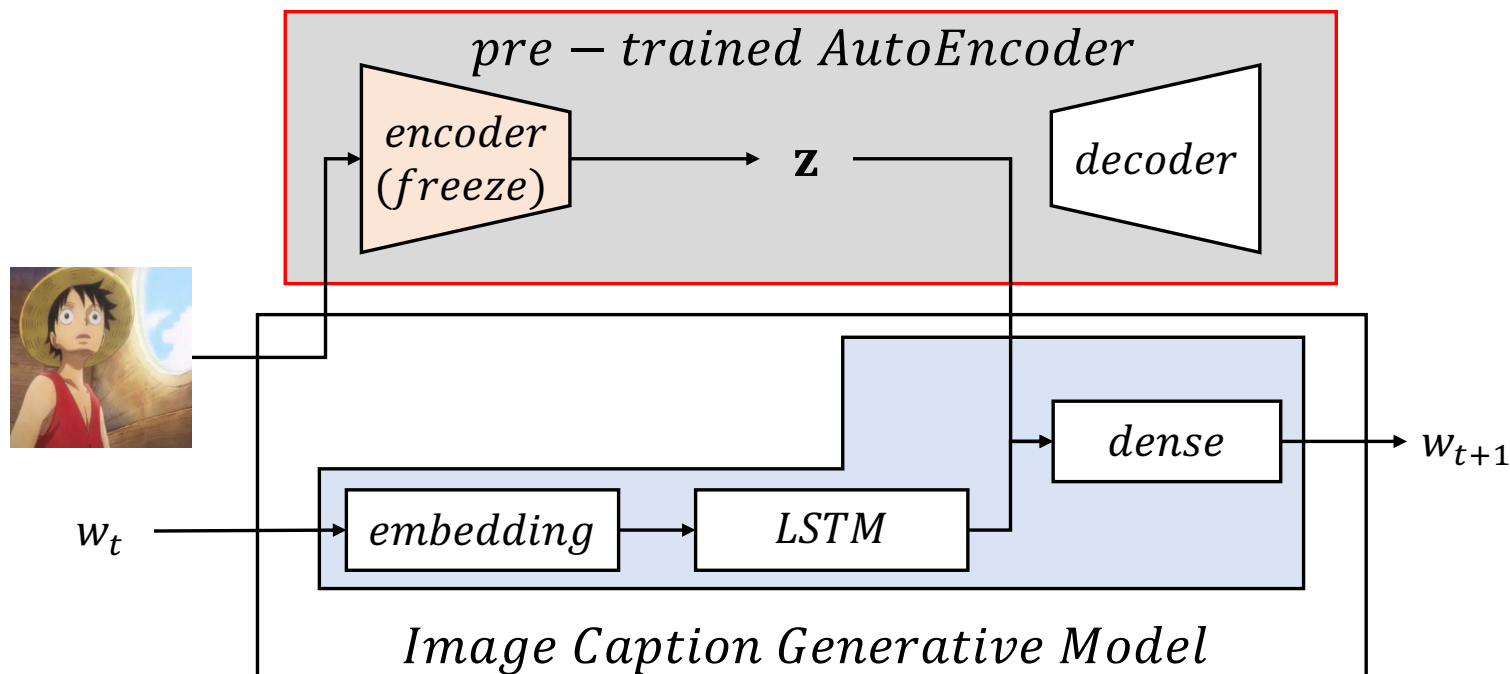


Fig. 8. The structure of GUMI-AE.

AutoEncoder[4]

- Unsupervised learning AI that learns to restore input.
- No need to annotate training data.
- Capable of dimensional compression feature extraction of data.

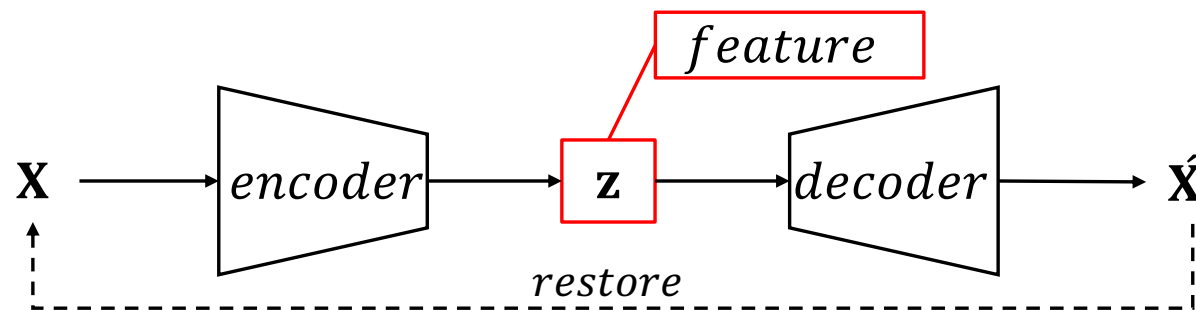


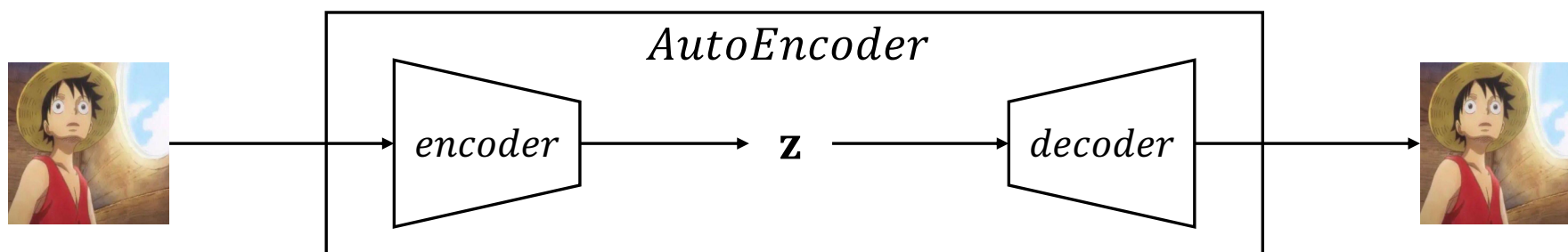
Fig. 9. Construct of AutoEncoder.

- AutoEncoder is suitable for unlabeled Bokete images.

[4] Hinton, G. E., and et al.: Reducing the dimensionality of data with neural networks. science, 313(5786), 504-507 (2006).

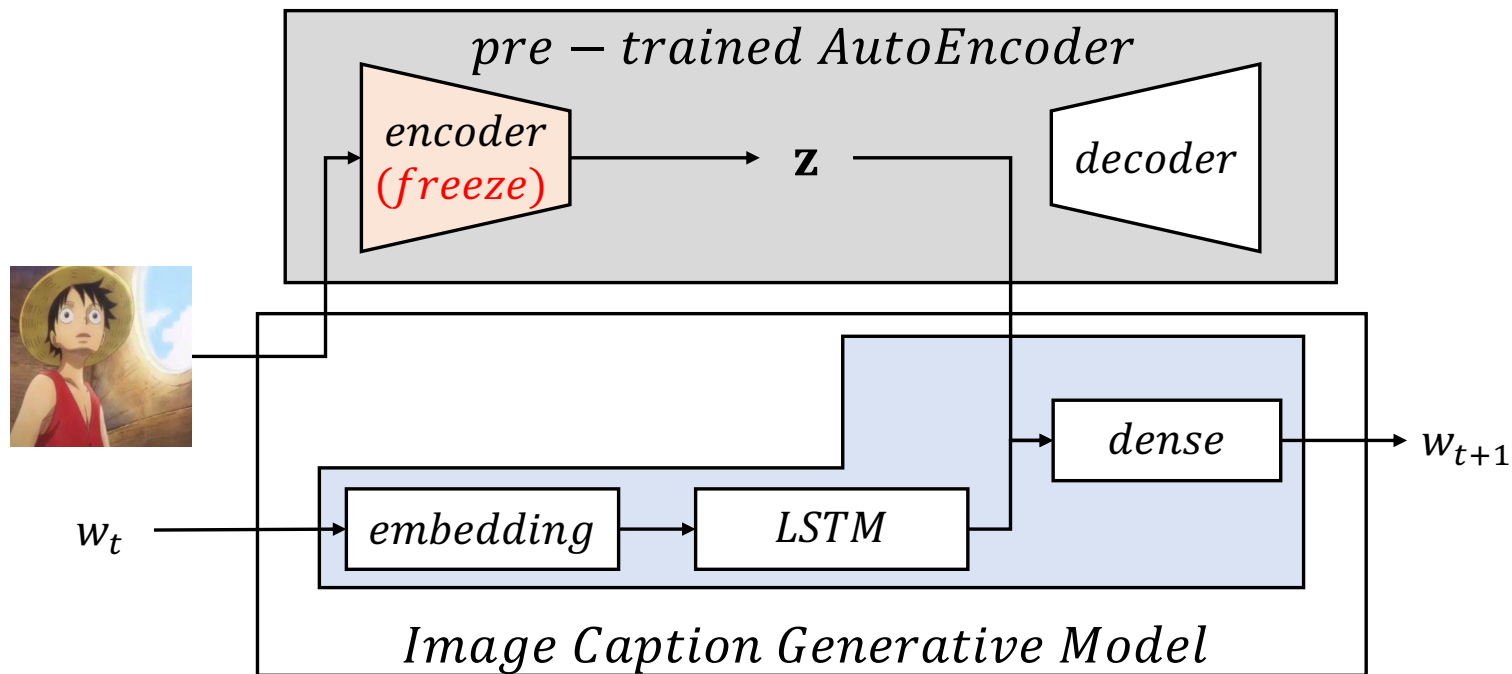
Proposed method : Step 1

Train an AutoEncoder using Bokete images.



Proposed method : Step 2

Freeze the parameters of the encoder and train image caption generative model.



Proposed method

Generative model of sUitable Meme sentences for Images using AutoEncoder; GUMI-AE.

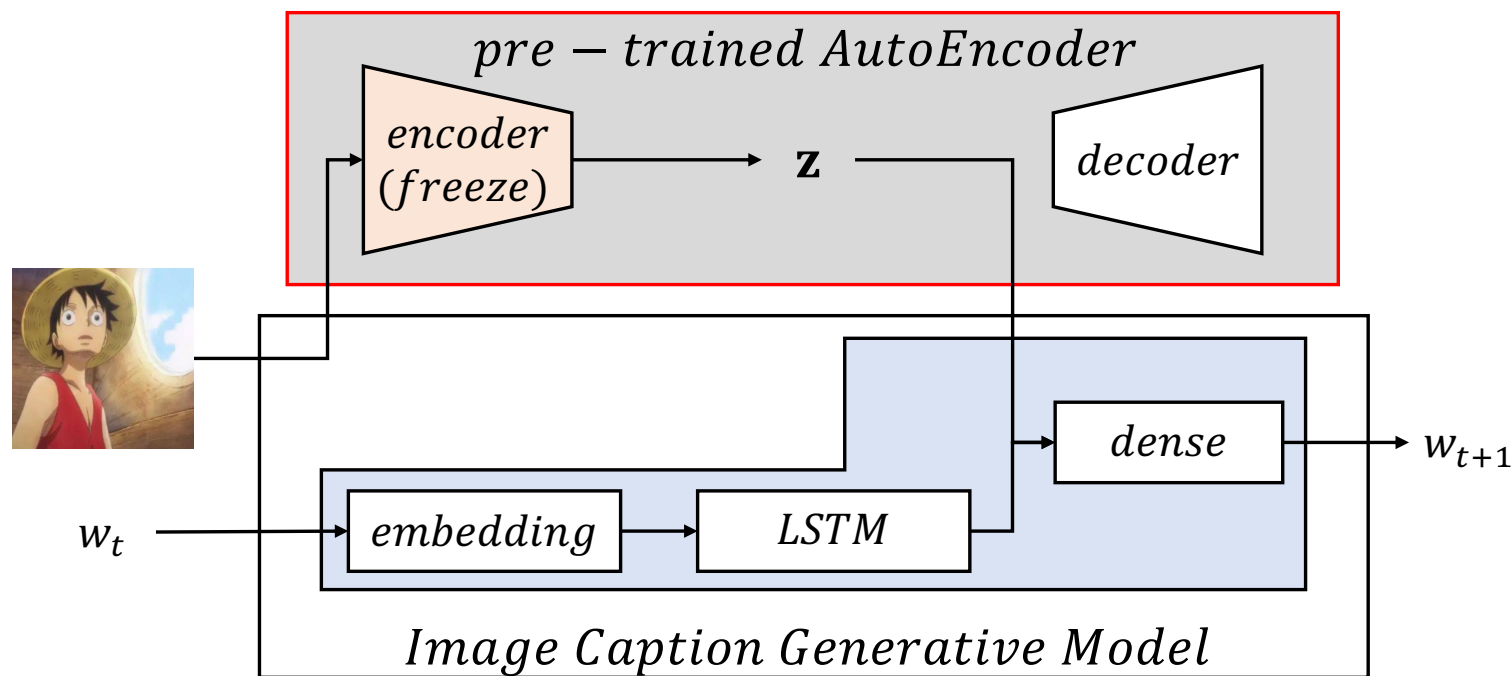


Fig. 8. The structure of GUMI-AE.

Comparison of GUMI-AE and NJM

Table 1. Comparison of GUMI-AE and NJM.

	GUMI-AE	Neural Joking Machine
pre-train	✓	×
Risk of domain divergence	Low	High
Flexibility of model	High	×

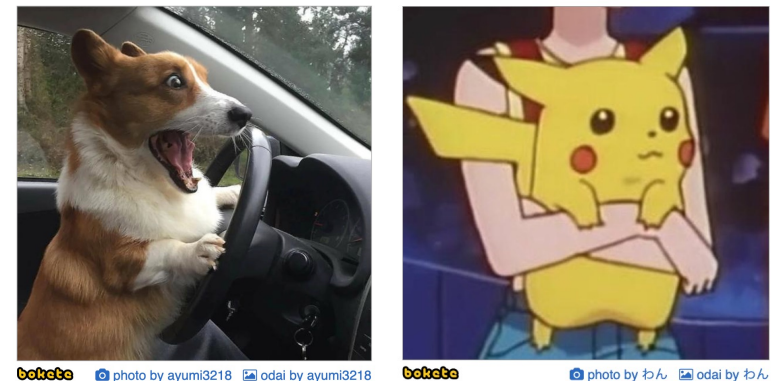
- GUMI-AE needs to be trained AutoEncoder in advance.
- GUMI-AE is suitable for Bokete images.
- The structure of GUMI-AE is defined freely.

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- **Experiments**
 - Dataset for training
 1. Training AutoEncoder
 2. Training meme generator
 3. Evaluation by questionnaire
- Conclusion

Dataset for training

- Scraping pairs of image and meme from Bokete website.
- Preprocess for dataset
 - Remove special characters such as “!” and “?” from memes.
 - Remove memes containing words that appear less than five times across the entire dataset.
- Amount of dataset
 - Image : 69,365
 - Memes : 552,613
 - * Multiple Meme's per image



猫が運転してた

AEDを持って来たわ!

Fig. 10. Memes posted in Bokete.

Experiment 1 : Training AutoEncoder

Change the output dimension of the encoder and verify its effect.

AutoEncoder uses a convolutional neural network.

- Optimizer : Adam($\eta = 0.001$)
- Minibatch size : 32
- Epochs : 150

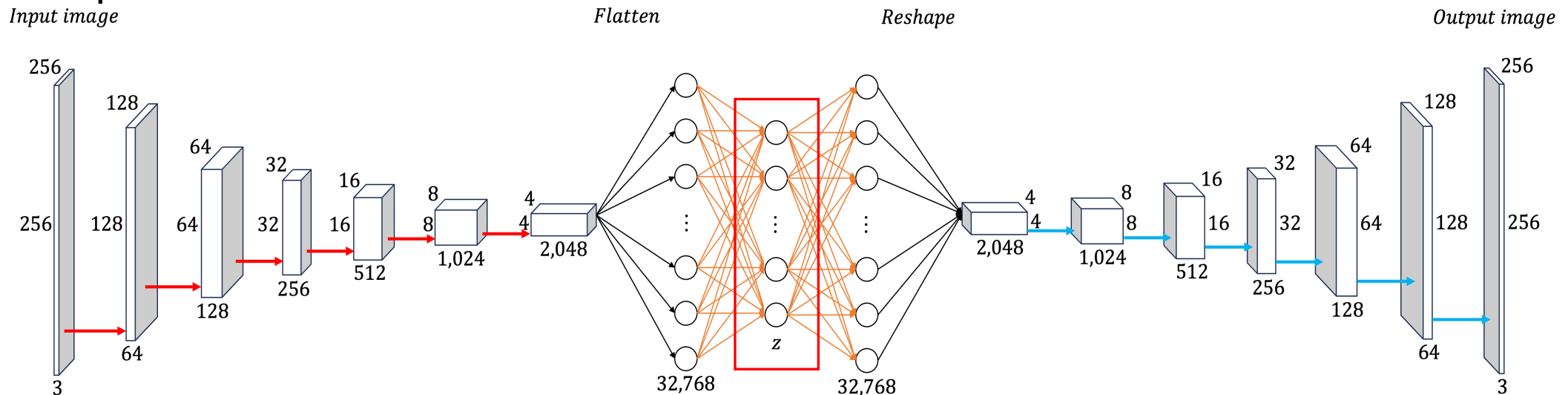


Fig. 11. The structure of the AutoEncoder.

Experiment 1 : Training AutoEncoder

FID[5] is a method of measuring the distance between two image groups.

- The closer to 0, the closer the two image groups are.
- Measuring the distance between input images and its corresponding output images.

Table 2. Result of experiment 1.

feature dim	512	1,024	2,048	4,096	8,192	16,384
loss	0.0065	0.0044	0.0031	0.0025	0.0020	0.0017
FID	246.04	187.39	142.02	130.20	124.51	109.69

- The increased output dimension of the encoder to have also increased the expressive power.

[5]Heusel, M, et al.: Gans trained by a two time-scale update rule converge to a local nash equilibrium. Advances in neural information processing systems 30 (2017).

Experiment 2 : Training meme generator

Train two types of meme generators.

1. GUMI-AE; using the encoder of the AutoEncoder trained previous experiment.
 2. NJM; using ResNet152, pre-trained on ImageNet.
- Optimizer : Adam($\eta = 0.001$)
 - Minibatch size : 2,048
 - Epochs : 100

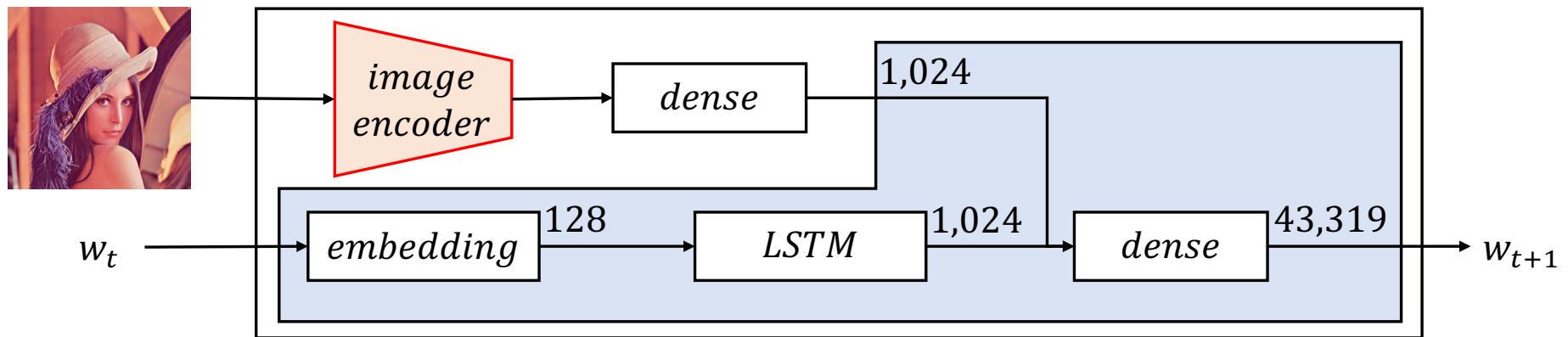


Fig. 12. The basic structure of the meme generator.

Experiment 2 : Training meme generator

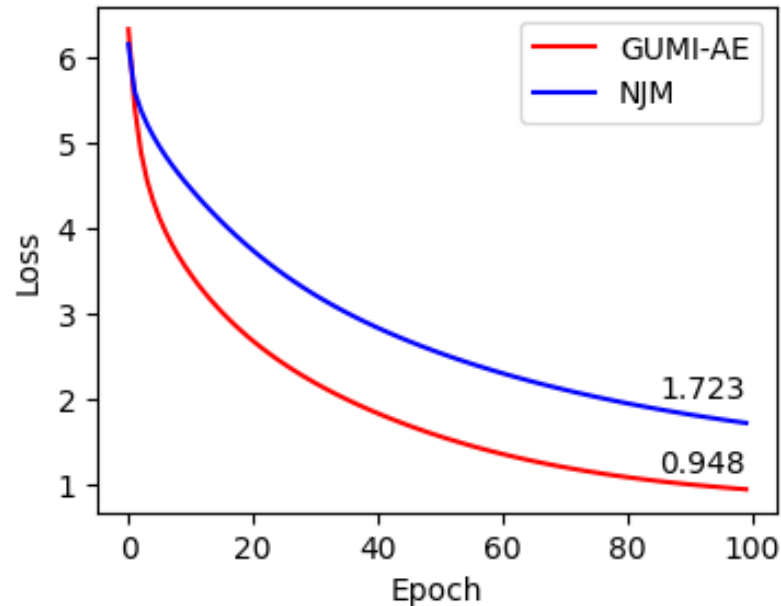


Fig. 13. Train loss per epoch.

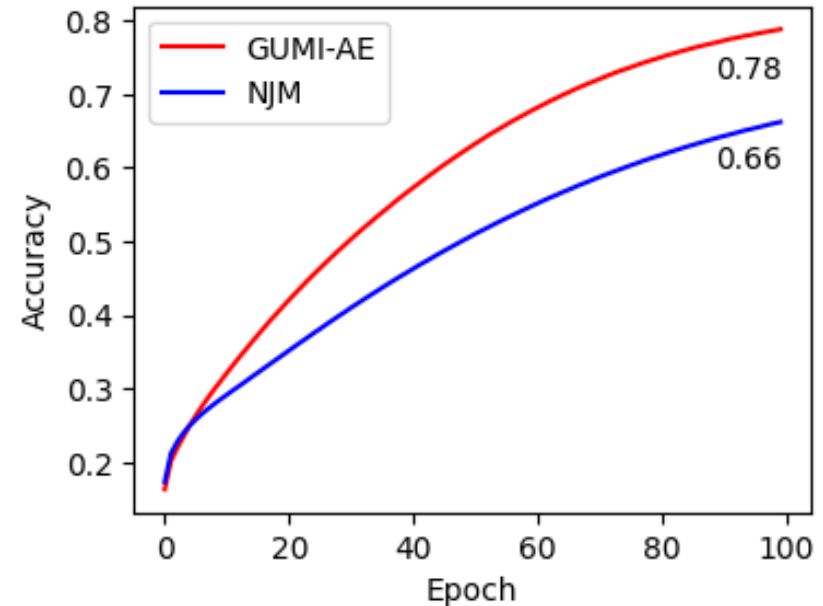


Fig. 14. Train accuracy per epoch.

- These performances difference are due to the encoder's ability to extract features.
- GUMI-AE achieved lower loss and higher accuracy than NJM.

Experiment 2 : Training meme generator

*These memes were translated from Japanese to English by DeepL.

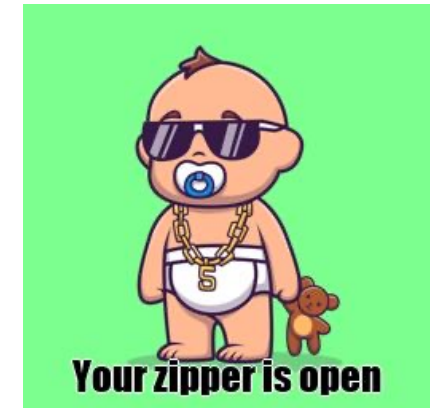
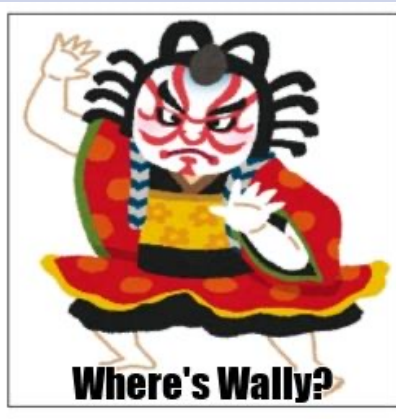
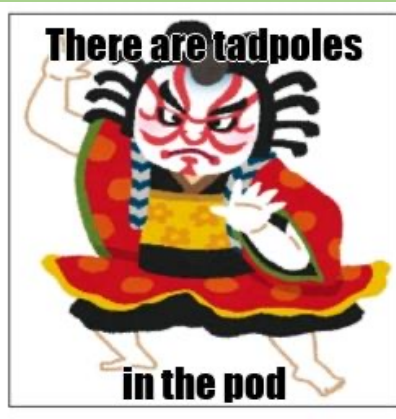
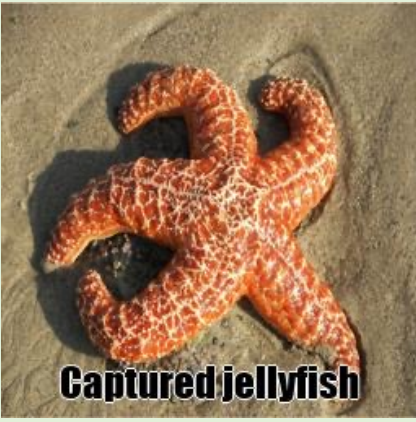
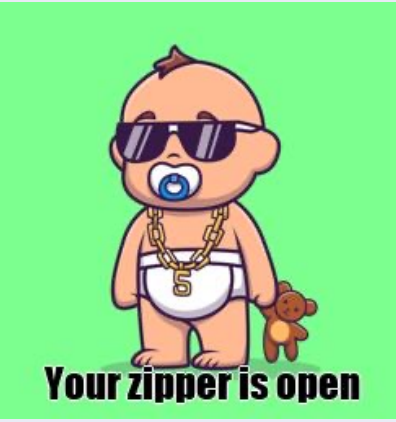


Fig. 15. Memes generated by GUMI-AE.

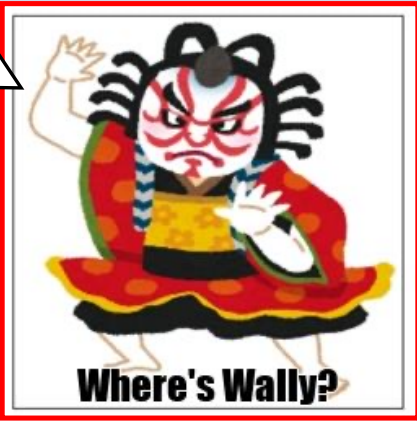
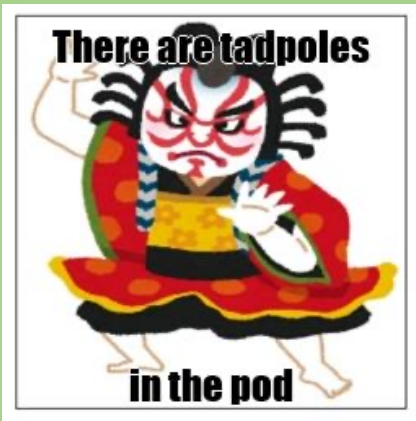
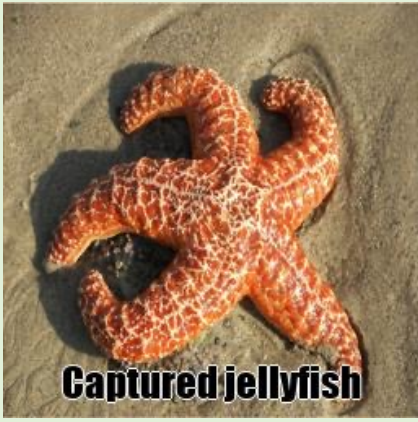
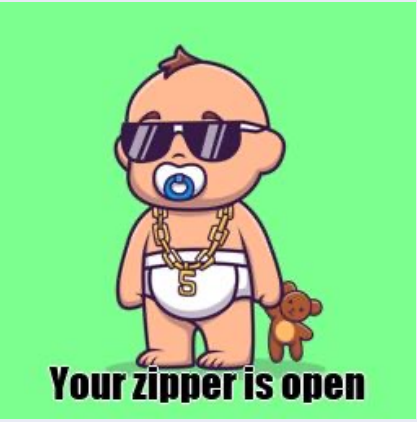
Experiment 2 : Training meme generator

Table 3. Comparison of memes generated by GUMI-AE and NJM.

GUMI-AE	Neural Joking Machine	GUMI-AE	Neural Joking Machine
 Can you take a walk?	 Ten minutes later, the bomb goes off.	 Where's Wally?	 There are tadpoles in the pod
 Perform one last gag	 Captured jellyfish	 Your zipper is open	 The sea will split

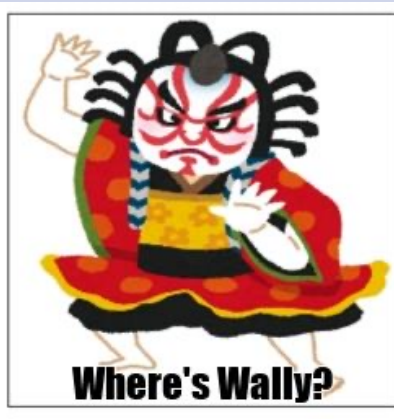
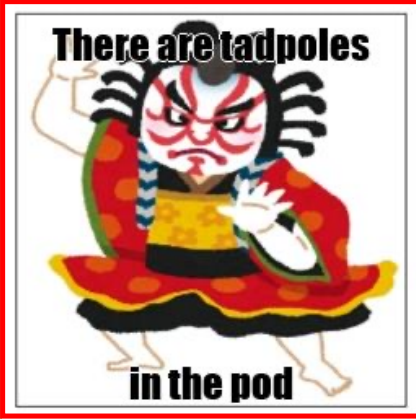
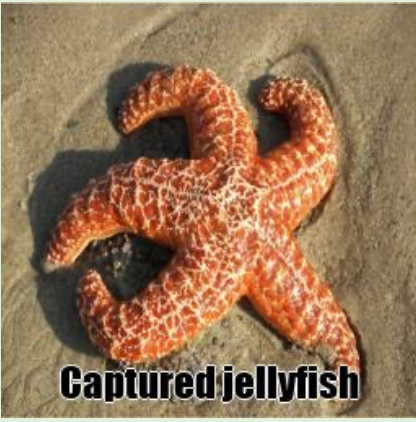
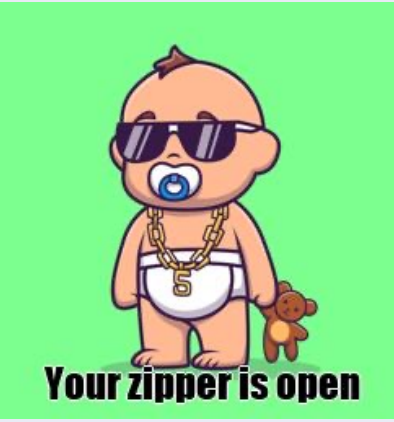
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Experiment 3 : Evaluation by questionnaire

There are no algorithms or methods to evaluate if a meme is fun.

Evaluate with a questionnaire if a meme is fun.

The questionnaire evaluated the fun of the meme in 4 level.

- 3. funny
- 2. not bad
- 1. bad
- 0. boring

The questionnaire was conducted anonymously using SNS.

Respondents must be residents of Japan.



Fig. 16. Interface of questionnaire.

Experiment 3 : Evaluation by questionnaire

*Total number of answers was 509.

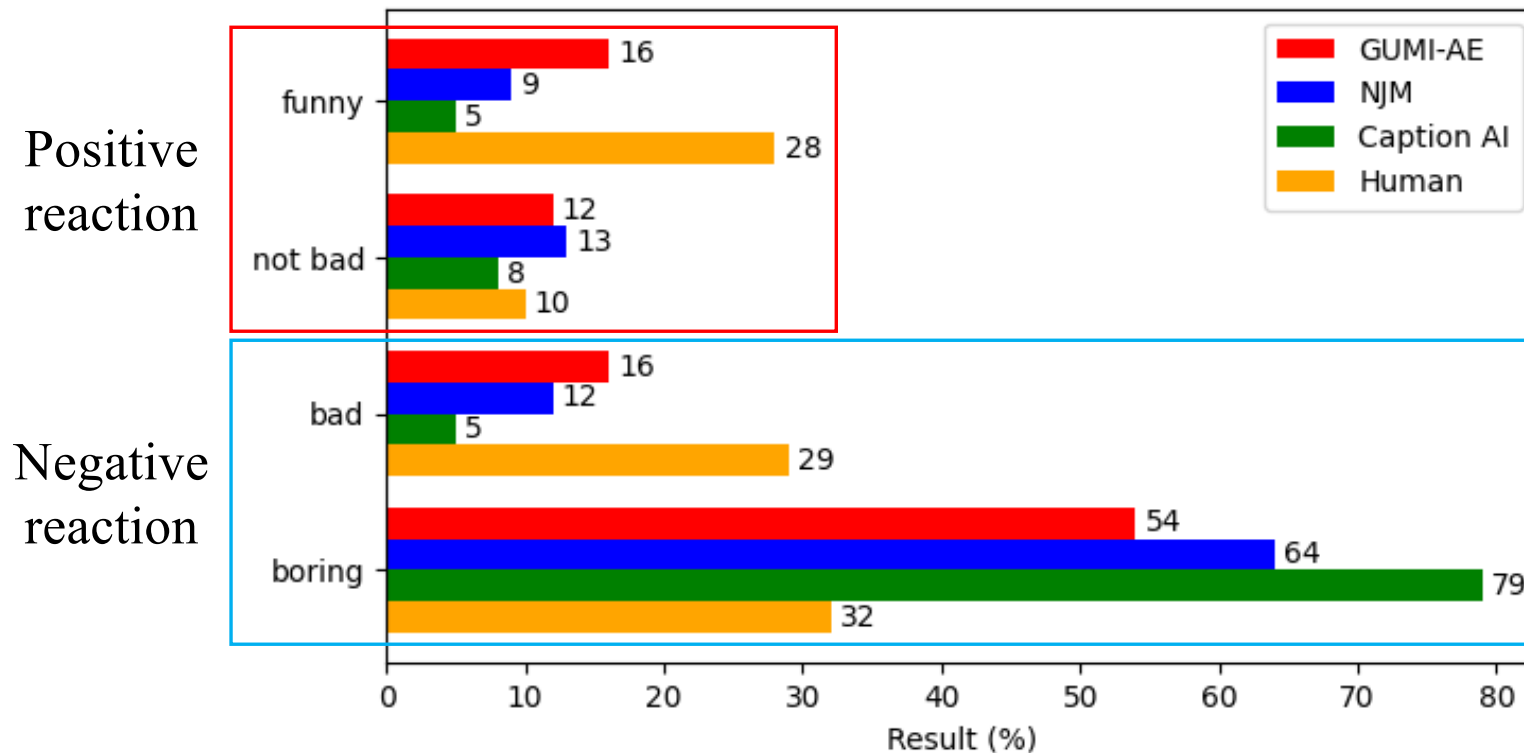


Fig. 17. Result of questionnaire.

- Roulette Selection might be generating grammatically incorrect memes.

Conclusion

- In this research, Propose a new generative model of meme using AutoEncoder, called GUMI-AE (Generative Model of Suitable Meme Sentences for Images using AutoEncoder).
- GUMI-AE dealt with domain divergence compared to conventional method NJM.
- Experiments showed that the implemented meme generator generates funny memes suitable for the images.
- According to the questionnaire, GUMI-AE generates funnier memes than NJM.

Future work

- Use advanced networks, such as Gated Recurrent Unit or Transformer.
- Implement the AI evaluate if a meme is funny or not for quantitated evaluation.
- Knowledge discovery from generative process of a meme generator to understand how memes are created.

Thank you for attention!

Any question? 

Image Caption Generative Model[6]

- CNN-based image encoder and LSTM-based sentence decoder was used.
- The model outputs the probability distribution of word recommendations and selects words by argmax or roulette selection.

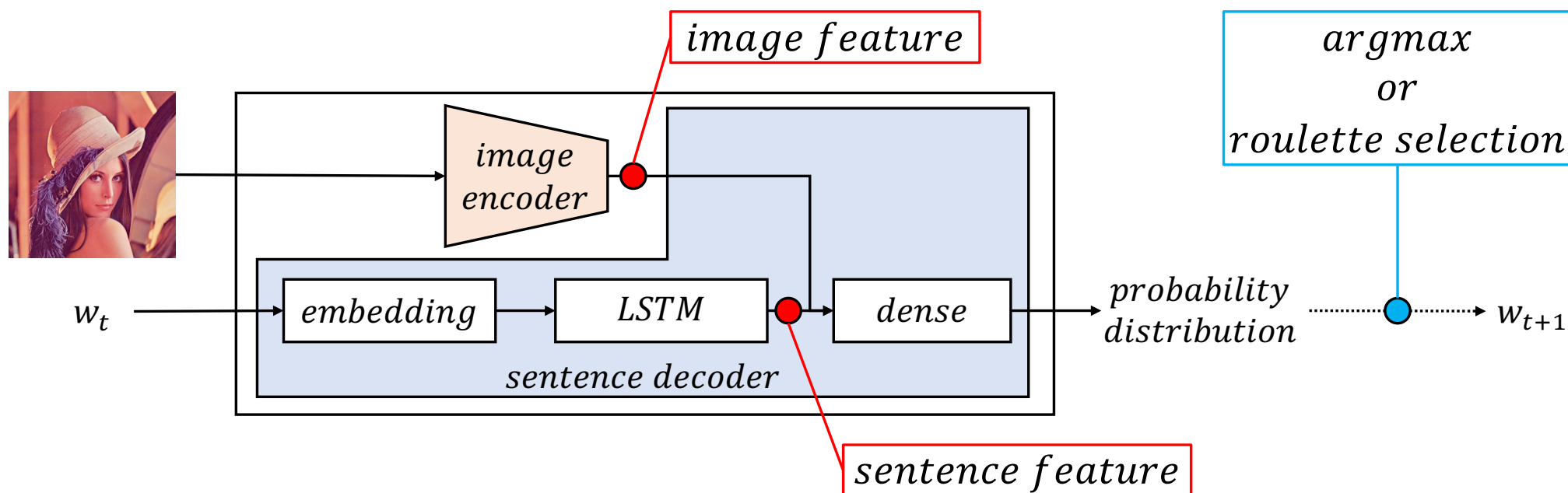


Fig. 18. The structure of image caption generative model.

[6] Vinyals, O., Toshev, A., Bengio, S., Erhan, D.: Show and tell: A neural image caption generator. Proc. IEEE Computer Vision and Pattern Recognition (CVPR), pp. 3156-3164 (2015).

Image Caption Generative Model

- Sentences are obtained by inputting words recursively into the model.

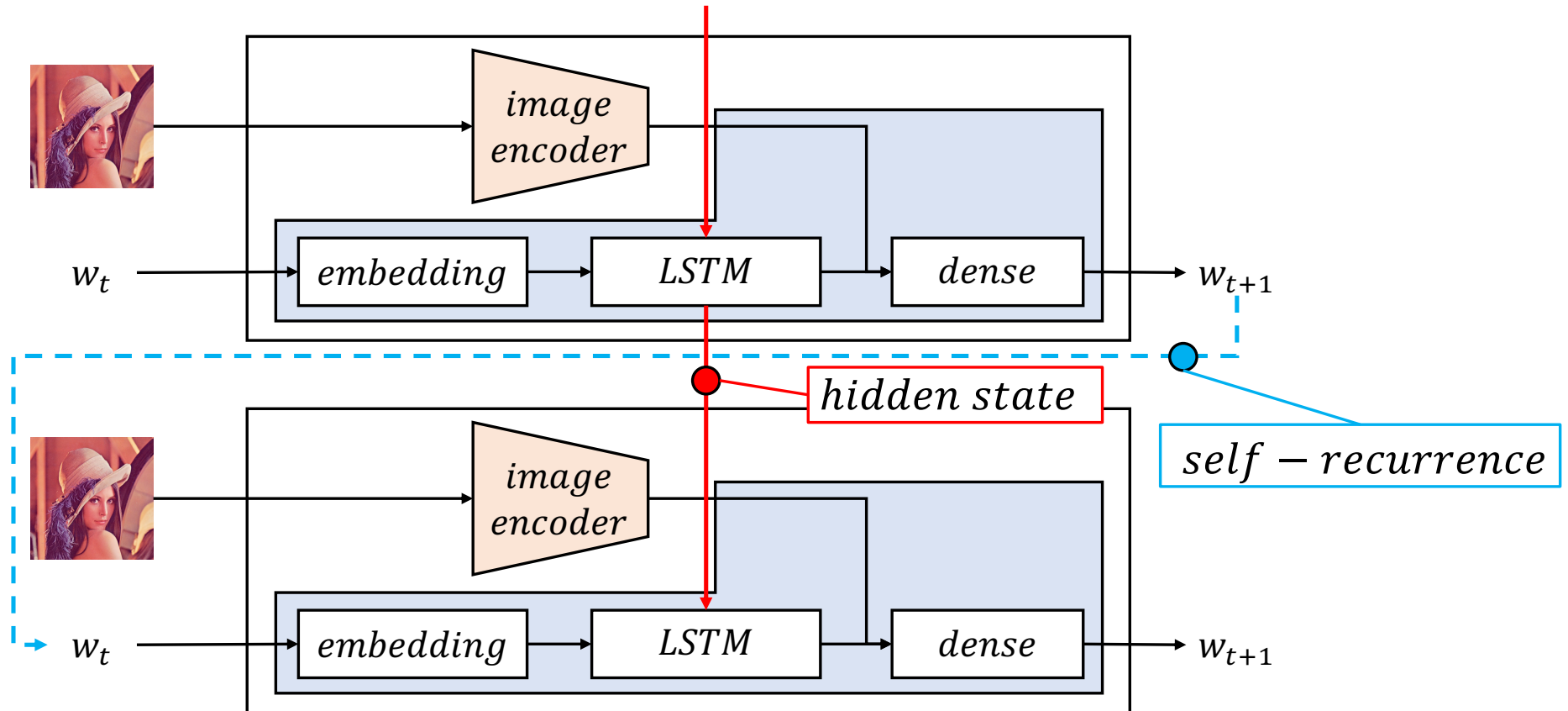


Fig. 18. The structure of image caption generative model.

3 methods to solve domain divergence

- Fine tuning

While learning image caption generation, the parameters of **the pre-trained model are also learned.**

- Transfer learning

While learning image caption generation, the parameters of **the pre-trained model are not also learned.**

- Relearn

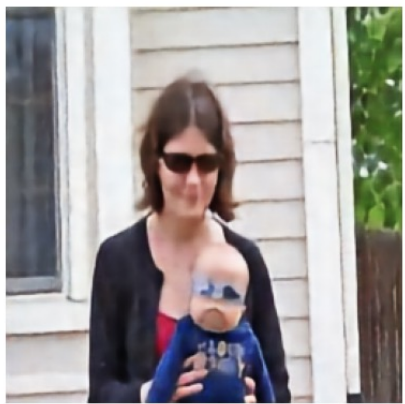
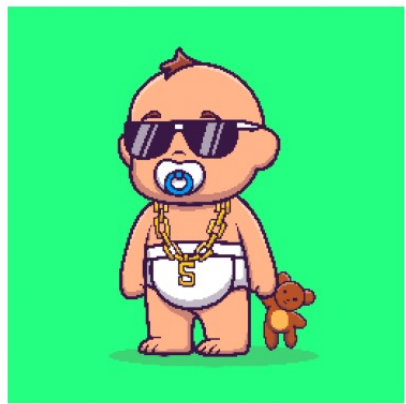
First relearn the pretrained model using target images, second freeze the parameters while learning image caption generation.

Table 4. Comparison of three methods.

	Fine tuning	Transfer learning	Relearn
Training cost	High	Low	Middle
Demand of label	-	-	○
Ability to solve domain divergence	High	Low	High
Risk of overlearning	High	Low	Middle

Experiment 1 : Training AutoEncoder

Table 5. Inputs and outputs of AutoEncoder trained in experiment 1.

Input(trained)	Output	Input(untrained)	Output
			
			

Experiment 2 : Training meme generator

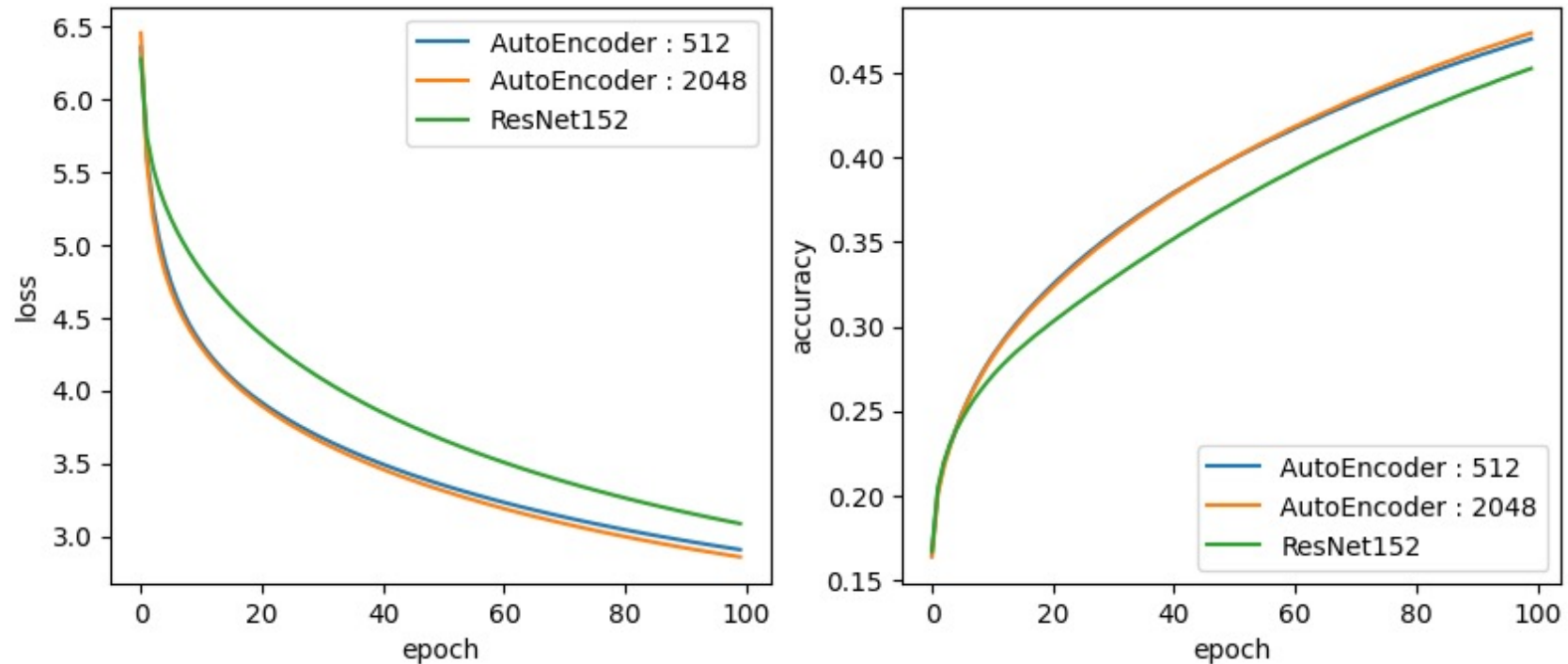


Fig. 19. Training loss / accuracy per epoch (lightweight model).

- AutoEncoder's encoder achieved lower training loss and higher training accuracy than the ResNet152 encoder

Experiment 3 : Evaluation by questionnaire

- Roulette selection was used for choosing words.
- Grammatically incorrect words were selected by Roulette selection.

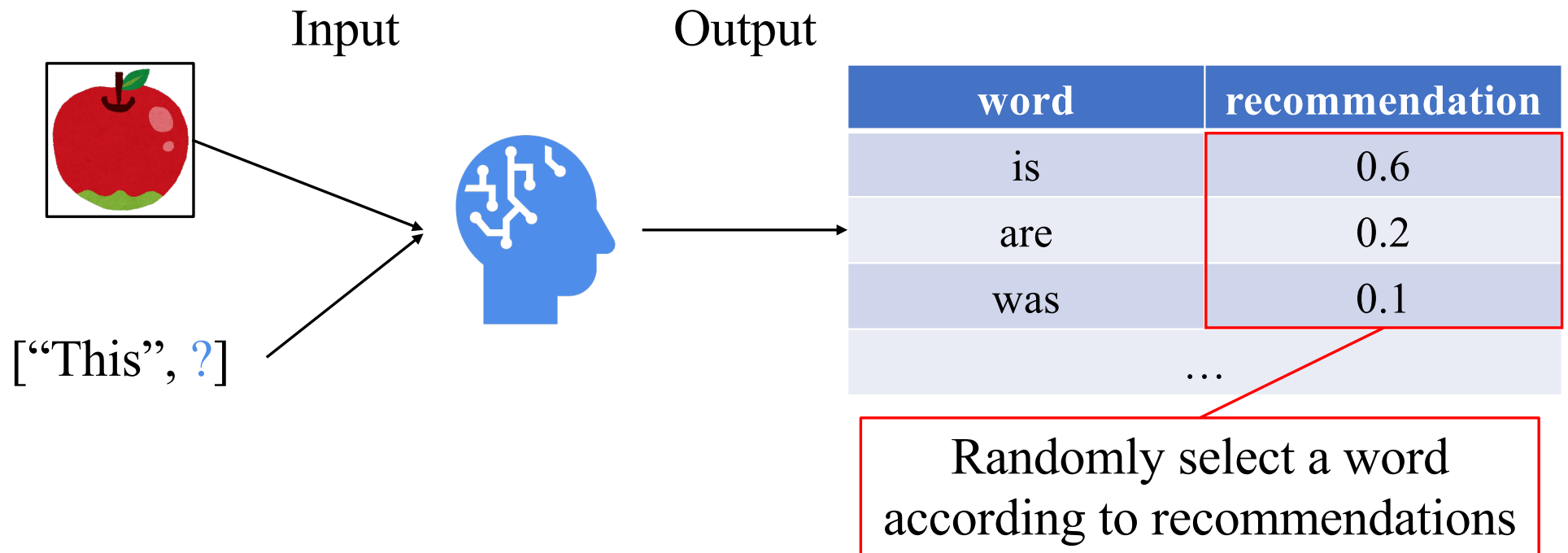


Fig. 20. Roulette selection.

Other Metrics

- BLUE, Perplexity

Word matching between the test and generated sentences.

In a ordinary image caption generation task would work well because the test and training data are similar.

In meme generation task would not well because the test and training data are not similar.

- BERT Score

Mean matching between the test and generated sentences.

This method depends on the BERT vocabulary

Bokete data has many words that are not in BERT's vocabulary, so it does not work.

GitHub

* Only Japanese model is available.



 <https://github.com/yryo1005/Neural-Ohgiri-Generator-GUMI-AE>